Large Language Models for Bayesian Network Parameterization in Software Engineering: Direct Elicitation versus Ranked Nodes

Anonymous Author(s)

Abstract

Software engineering (SE) decision makers need models they can understand and use when data are scarce. Bayesian networks (BNs) address this by making probabilistic relationships explicit and incorporating expert knowledge, but eliciting their conditional probability tables (CPTs) is costly. The Ranked Nodes Method (RNM) generates ordinal CPTs from fewer judgments. Should large language models (LLMs) fill complete CPTs or supply fewer judgments to RNM? We compared complete CPTs for 61 ordinal conditional nodes from seven SE BNs in an exploratory study. We tested structured prompts that clarified conditioning, framed probabilities as hypothetical frequencies, and requested consistency checks. Qualitative retrieval-augmented generation (RAG) supplied short passages on domain relationships, without reference probabilities or numerical parameters. Eight conditions used GPT-6 Astra at medium effort across five replications. With equal node weights, total variation averaged 0.2317 for direct elicitation and 0.2404 for RNM, a 3.76% increase. Because nodes were unevenly distributed across five source studies, we weighted each study equally; direct elicitation still had lower error. Both contrasts passed Holm-adjusted paired t-tests, but not exact sign tests. RNM improved 31 nodes, yet larger deteriorations elsewhere outweighed those gains. Structured prompting modestly helped direct elicitation under node weighting; qualitative RAG demonstrated no benefit. For these small CPTs, RNM requested 40.1% fewer distributions and used 12.1% fewer output tokens, but only 0.61% fewer total tokens. The findings favor direct elicitation as the average numerical baseline in this sample and show why reduced elicitation alone cannot establish a better, cheaper-to-complete CPT.

CCS Concepts

• **Computing methodologies** → **Bayesian networks**; • **Software and its engineering** → *Software creation and management*.

Keywords

Bayesian networks, large language models, probability elicitation, ranked nodes, software engineering

1 Introduction

Supporting software engineering (SE) decisions faces two main challenges: explaining the basis for a recommendation and obtaining enough relevant data to build a model experts can rely on. Examples include release readiness, quality assessment, and process improvement, where decision makers must judge whether a recommendation's assumptions apply. A review of 108 studies documents the demand for explainability across 23 SE tasks [3]. Training data for machine learning and deep learning may also be scarce, particularly for managerial SE decisions [23]. Explicit assumptions and domain knowledge are therefore valuable.

Bayesian networks (BNs) offer such an approach. A BN makes relationships and uncertainty explicit in a graph and probabilities, incorporating domain knowledge when data are limited [10, 11]. BNs have supported many SE problems, including early defect prediction [10], effort estimation [17], and diagnosis of development-process problems [21]. A systematic mapping of 106 studies identifies software management and quality as prominent application areas [24]. Building a credible BN nevertheless requires substantial knowledge engineering effort.

Constructing conditional probability tables (CPTs) is a major part of that effort. Each row describes a child's probabilities for one combination of its parents' states; adding parents rapidly increases the number of judgments required [11]. Semi-automatic methods reduce this burden by reconstructing a table from fewer expert judgments or parameters. These include the Ranked Nodes Method (RNM) [11], the Weighted Sum Algorithm (WSA) [20], Analytic Hierarchy Process (AHP)-based pairwise comparisons [5], and interpolation methods [1, 2]. Their assumptions can, however, limit fidelity to the intended probabilities [2].

Large language models (LLMs) are now being explored for CPT construction. Nafar et al. elicited numerical BN parameters directly [18]; Gottal and Matthes reconstructed CPTs from statistical summaries extracted from scientific literature [12]; and Cypko et al. used retrieval-augmented generation (RAG) to support expert BN modeling [6]. Should an LLM specify every row, or supply fewer judgments for a semi-automatic method to complete? Any benefit depends on both the judgments and the reconstruction.

We investigated this question using RNM, which applies to variables with ordered states such as Low, Medium, and High [11, 15]. Instead of requesting the entire CPT, we asked the LLM for distributions at selected parent-state combinations, called *anchors*, and for a rule describing how the parents combine. A numerical fitting procedure then adjusted that rule's weights and uncertainty to match the elicited distributions and generated the complete CPT. We compared this approach with direct complete-table elicitation on 61 ordinal conditional nodes from seven SE BNs.

We also examined support for elicitation. Structured prompts made conditioning explicit and asked for probability and consistency checks; qualitative RAG supplied short passages explaining relevant domain relationships. Crossing the two elicitation methods, two prompt designs, and presence or absence of RAG produced eight conditions. We repeated each five times using GPT-6 Astra at medium effort. This study addresses three research questions:

- **RQ1—Reference agreement:** How does LLM-assisted RNM compare with direct elicitation in complete-CPT agreement across the full SE sample?

SAC '27, Gwangju, South Korea
2027.

- **RQ2—Elicitation support:** How do structured prompting and qualitative RAG affect each method's agreement across complete tables?
- **RQ3—Accuracy and effort:** How does RNM trade reference agreement against requested judgments and observed computational effort?

All three RQs include all 61 nodes and every CPT row, including anchors. The seven BNs cover Scrum process assessment [19], Chrome code review [14], ERP data quality [13], software maintenance logs [8], and software strategic indicators [16]. We give nodes equal weight for the main comparison. Because BNs contribute unequal numbers of nodes, we also weight their five source studies equally and report each BN separately. This evaluates the complete tables available for use rather than restricting attention to configurations omitted from RNM prompts.

Section 2 positions the study in related work, Section 3 describes the experiment, and Section 4 reports the results. Section 5 discusses the RQs and implications for SE, Section 6 examines validity threats, and Section 7 concludes the paper.

2 Background and Related Work

This section explains RNM and briefly introduces alternative CPT construction techniques (Section 2.1), then examines LLM approaches for CPT construction and positions our contributions (Section 2.2).

2.1 Ranked Nodes Method

RNM generates CPTs for ordinal variables such as Low, Medium, and High from a compact description of parent influence and uncertainty [11, 15]. Ordered states correspond to intervals on a common $[0, 1]$ scale. For example, three equally spaced states occupy successive thirds of that scale. For each parent-state configuration, a weighted function combines values from the corresponding parent intervals. A normal distribution, truncated to $[0, 1]$ and centered at the function value before truncation, assigns probability mass to the child's state intervals. Averaging over parent values produces the CPT row.

The function, weights, and variance express different judgments. A weighted mean allows stronger parents to compensate for weaker ones; weighted-minimum and weighted-maximum functions emphasize weaker or stronger parent values. Weights adjust parent influence, while variance controls dispersion across child states. Thus experts specify a relationship and uncertainty rather than independently assessing every row. Parameters can also be fitted to selected elicited distributions, as in our study. The same settings generate all rows, reducing elicitation but restricting the relationships representable. Laitila and Virtanen show that RNM can approximate many CPT relationships while remaining sensitive to parameter choices [15].

RNM is one of several semi-automatic approaches. WSA combines elicited distributions through weighted sums [20]; AHP uses pairwise comparisons [5]; interpolation methods, including InterBeta, infer unassessed distributions from selected inputs [1, 2]. These alternatives use different judgments and assumptions, so reduced elicitation alone does not establish numerical fidelity.

2.2 LLMs for CPT construction

LLM approaches differ in elicitation granularity and supplied evidence. Nafar et al. found that requesting one child-state probability at a time lowered median BN KL divergence for seven of eight models, with more queries [18]. Their FullDist baseline requests one conditional row; ours requests the whole CPT. Their result leaves open whether eliciting fewer configurations and reconstructing the rest improves agreement. Gottal and Matthes reconstruct from numerical summaries [12], which constrain probabilities more directly than qualitative RAG. Rahul and Chowdhury's persona-panel preprint evaluates effort and internal consistency, without independent reference-CPT accuracy [22].

Other work evaluates modeling or predictive support. BIRD illustrates neural–symbolic integration [7] through LLM judgments, constrained fitting, and BN inference [9]. Clinical RAG assists BN modelers [6]; AutoElicit evaluates prediction after updating elicited parameter priors [4]. These outcomes complement numerical CPT agreement but cannot establish it.

We connect these research directions by comparing direct elicitation with LLM anchors plus RNM fitting across complete CPTs for 61 nodes from seven SE BNs. We examine structured prompting, qualitative RAG, and token usage, with anchor diagnostics separating elicitation from fitting. Experimental fits use LLM judgments without reference access. Table 1 compares the studies by supplied evidence, evaluation endpoint, and scope.

3 Method

Figure 1 follows the sample through eight conditions crossing elicitation method, prompt style, and qualitative RAG, each repeated five times. In the main experiment, references entered only at evaluation; both methods produced complete CPTs for all 61 nodes.

3.1 BNs and reference CPTs

The sample contains seven BNs from five SE studies. Scrum assesses team and process factors affecting sprint and backlog quality [19]; Chrome 40 and 42 concern code review and post-release defects [14]; APS concerns ERP data quality [13]; SESSI concerns software strategic indicators [16]; and two BNs concern maintenance logs [8]. Versions from one study form a publication family.

We selected conditional nodes with complete numerical references and interpretable, ordered child and parent states. Binary variables qualified when their meanings supported an order; nominal variables and Boolean gate chains were excluded. Of the 61 CPTs, 36 have at most four rows, 20 have six or nine, and five have 27; all five largest are Scrum. In 31 nodes, anchors cover every row. Scrum uses an RNM-derived reference; the other BNs supply tabulated probabilities. References describe particular contexts, not universal company probabilities; versions and preparation are in the supplement.

All CPT rows, single-parent nodes, and fully elicited nodes remain included: fitting can change even anchor probabilities. Sampling is availability-based; the endpoint, weighting, and tests followed inspection of earlier results, making this analysis exploratory.

Table 1: LLM-assisted BN construction compared by inputs, evaluation, and scope. Entries distinguish numerical CPT agreement from modeling support and downstream performance; reported errors are not directly comparable. KL denotes Kullback–Leibler divergence.

Study	Construction and supplied evidence	Evaluation material and endpoint	Scope relative to our comparison
Nafar et al. [18] TMLR, 2026	LLM internal knowledge; one conditional row or one child-state probability per query. A separate stage combines estimates with observations.	80 public BNs, eight LLMs: BN/CPT KL. Separate classification evaluation on nine datasets with three LLMs.	Changes granularity within a row; our RNM approach requests selected rows and reconstructs the rest.
Gottal and Matthes [12] Findings ACL, 2026	Reconstruction algorithms and direct LLM CPT output from numerical statistical summaries; separate extraction from publications.	Synthetic BNs and publication/raw-data cases; CPT distances, inference and decision utility, and runtime.	Quantitative summaries constrain construction; our RAG packets supply qualitative context.
Rahul and Chowdhury [22] Preprint, 2026	Persona-conditioned LLM panels provide distributions for parent configurations; trimmed-mean aggregation.	One healthcare-consultation BN; plausibility and internal consistency checks; calls, token estimates, and runtime.	Demonstrates construction feasibility and effort; lacks an independent numerical CPT benchmark.
Feng et al. (BIRD) [9] ICLR, 2025	LLM-generated factors and coarse likelihood judgments feed constrained parameter fitting and BN inference.	Human judgments of relative support and downstream task accuracy.	Establishes a precedent for LLM judgments plus fitting; does not test numerical reference-CPT recovery.
Cypko et al. [6] IJCARS, 2026	Clinical RAG supports expert BN modeling using guidelines and textbooks.	Synthetic retrieval evaluation; modeling, usability, and workload study with four clinicians.	Tests modeling support rather than complete numerical CPT agreement.
This study	Complete CPT elicitation versus LLM anchors plus RNM fitting; baseline/structured prompts, with/without qualitative RAG.	61 ordinal conditional nodes from seven SE BNs; complete-table distances, five replications, tokens, and fitting time.	Compares final CPTs on all rows of all nodes; checks weighting sensitivity and uses anchor fitting as a diagnostic.

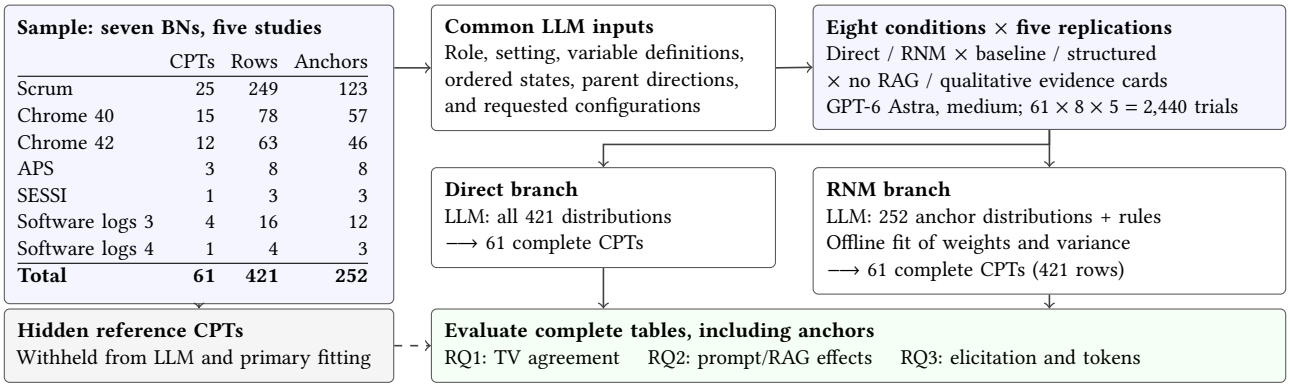


Figure 1: Main experiment: one CPT per selected node. Rows and anchors are counted per replication and prompt/evidence combination. Solid arrows show elicitation and processing; dashed arrows carry hidden references only to evaluation. RNM fits LLM anchors.

3.2 Inputs and outputs in the eight conditions

Common inputs. Answers conditioned on all listed parents, averaging over unspecified circumstances. Graphs, definitions, prompts, and numerical procedures were fixed before collection.

Methods and simplified prompts. Cohesion depends on Self-Organizing and Collaboration; all use Low, Medium, and High states. These simplified excerpts illustrate the branches: **Direct:** Estimate Low, Medium, and High Cohesion probabilities for all nine combinations of Self-Organizing and Collaboration. Return normalized distributions in JSON, preserving state order.

RNM: Return the five listed anchor distributions. Select a weighted mean, weighted minimum, weighted maximum, or minimum–maximum rule. Weights and variance will be fitted offline to generate remaining rows.

Both methods returned vectors and an evidence note; only RNM enforced influence assumptions mathematically. Full prompts accompany the data.

Prompt and evidence additions. Baseline prompts requested distributions directly; structured prompts added: **Structured addition:** For each requested configuration, consider 100 hypothetical comparable cases. Estimate the child given the parents, distinguish evidence from assumptions, and check that probabilities sum to one.

Hypothetical cases supplied no observations; fractional counts were allowed. RAG added qualitative cards. Neither addition changed the requested rows or output format.

RAG matched curated domain paraphrases to model context and target/parent tags, selecting up to three cards (24–103 words; median 72). Cohesion received cards on shared goals, organizing work, and knowledge coverage, including:

> **Shared goals.** Cohesive teamwork includes commitment to shared tasks and team goals. Agile collaboration involves joint responsibility, rather than treating every task as an isolated individual obligation.

Packets were reused across methods, styles, and replications, excluding probabilities, parameters, and identifying links. No-RAG requests retained definitions without cards. RAG tested qualitative context, not live retrieval or company measurements; input budgets were not equalized.

3.3 Producing comparable complete tables

Direct outputs were complete CPTs. RNM fitted weights and variance to the selected rule and anchor distributions, then generated all rows [11, 15]. It minimized mean anchor TV without accessing reference probabilities; valid poor fits were retained. The supplement details equations, settings, and influence directions.

Selecting anchors. Anchors were parent configurations chosen deterministically, without reference probabilities or LLM answers. Using influence-aligned ordinal ranks, we selected all-low, all-high, and all-central configurations. For each parent, we also selected one-high/rest-low and one-low/rest-high, then removed duplicates. With even cardinalities, we used the lower central category, i.e., the lower influence rank for binary parents.

Example: Cohesion. The ordered pair is (Self-Organizing, Collaboration), with Low (L), Medium (M), and High (H). Its nine-row CPT yields five anchors: (L, L) and (H, H) capture unanimous extremes; (M, M) is central; (H, L) and (L, H) capture opposing extremes. Both one-parent rules produce the same opposing pairs, counted once. The LLM supplies a three-probability Cohesion vector for each anchor; it does not choose the rows. RNM leaves (L, M), (M, L), (M, H), and (H, M) unqueried, then fits and generates all nine rows, including anchors. Direct elicitation requests all nine.

For k parents with cardinalities r_i and a child with c states, a CPT has $N = \prod_i r_i$ rows and cN probabilities. With ternary parents, our anchor rule gives $A = 3$ for $k = 1$, $A = 5$ for $k = 2$, and $A = 2k+3$ for $k \geq 3$. Thus $1-A/N$ is the exact reduction in requested distributions, conditional on retaining this rule.

3.4 Execution and reference isolation

Collection used Codex CLI 0.162.1 and `gpt-6-astra` at `medium` effort on 15–16 October 2026. Requests used fresh sessions and were shuffled within each replication; temperature and generation seed were unspecified. Experimental sessions could not access references, full papers, evaluator code, or authoring conversations; tools, browsing, connectors, and memory were disabled.

Outputs required correct rows/states, finite probabilities in $[0, 1]$, and sums within 10^{-6} of one. One fresh-session format repair was allowed without reference feedback. All 2,440 trials produced valid outputs after four repairs (two per method); accounting includes all 2,444 attempts, with no fallback distributions.

3.5 Outcomes and analysis

For reference p and prediction q, lower TV means closer agreement:

$$\mathrm{TV}(p, q) = \tfrac{1}{2} \sum_s |p_s - q_s| \in [0, 1]. \quad (1)$$

We average TV over rows within each node, then give all 61 nodes equal weight, preventing larger CPTs from dominating. Within each replication, RQ1 averages RNM-minus-direct differences over four prompt/evidence combinations. RQ2 contrasts structured minus baseline and RAG minus no RAG within each method, averaging the other factor.

Scrum and Chrome supply 52 of 61 nodes, dominating the node-weighted mean. A sensitivity analysis gives each of the five publication families equal weight, averaging nodes within BNs and BNs within families. BN results are descriptive; supplementary Jensen–Shannon, ordinal-CDF, and smoothed CPT KL distances use the same weights.

RQ3 counts requested distributions before repairs and client-reported tokens over all attempts for the full sample. Total tokens equal input plus output; cached-input and reasoning-output counters are subsets, not additional tokens. Paired contrasts use replication totals. Latency and offline fitting time are descriptive; subscription tokens are not converted into monetary savings.

The five paired replication blocks are the inferential units, not correlated nodes or rows. Two-sided paired t-tests ($df = 4$) and pointwise 95% intervals assume approximately normal, independent, stationary block differences. Two-sided exact sign tests assess direction/median sensitivity under independence. Holm adjustments cover two RQ1 weighting contrasts, eight RQ2 contrasts (two methods, two factors, two weighting rules), and two RQ3 token contrasts. These are separate exploratory families, not study-wide confirmatory control; intervals are not simultaneous. No practical-effect or equivalence margin was prespecified.

3.6 Supporting analyses

An exploratory scaling analysis separates exact elicitation counts from conditional token projections. For the 45 nodes with ternary parents and child, we fitted separate ordinary least-squares models per method: mean total tokens as an intercept plus a coefficient times requested rows. Each node contributed its mean over 20 trials per method, including repairs. Scenarios above 27 rows extrapolate beyond observations. The supplement gives coefficients, a held-out-size check, and restricted fitting diagnostics; reference-informed fits assess best-found approximation, not error floors.

4 Results

We first compare the accuracy of the completed CPTs (RQ1), then ask whether prompting and RAG improve that accuracy (RQ2), and

Table 2: Complete-table TV by BN, averaged over prompt/evidence conditions and replications. Δ is RNM minus direct. Every node is included; BN rows are descriptive.

BN	Nodes	Direct	RNM	Δ
Scrum	25	0.2605	0.2334	−0.0272
Chrome 40	15	0.2069	0.2220	+0.0151
Chrome 42	12	0.1988	0.2335	+0.0347
APS	3	0.0964	0.1155	+0.0191
SESSI	1	0.1160	0.1143	−0.0017
Software logs 3	4	0.3802	0.4770	+0.0967
Software logs 4	1	0.2052	0.3299	+0.1247
All nodes	61	0.2317	0.2404	+0.0087

finally examine the effort saved by eliciting fewer rows (RQ3). All main results include every row of all 61 nodes. Lower TV means closer reference agreement; negative RNM-minus-direct differences favor RNM.

4.1 RQ1: Which method produces closer complete CPTs?

Direct elicitation is closer overall. Mean TV is 0.2317 for direct elicitation and 0.2404 for RNM, averaging each method's four combinations of prompt and evidence and five replications. RNM increases error by 0.0087, or 3.76%. All five replication differences are positive (0.0072–0.0094), with pointwise 95% t interval $[0.0076, 0.0098]$ for their mean. The Holm-adjusted paired-t test rejects ($p = 0.000048$), but the exact sign test does not ($p = 0.125$). The direction is consistent across runs, with limited distribution-free evidence; neither test establishes practical importance.

Scrum and Chrome supply 52 of 61 nodes. We therefore also give each source study equal weight. Direct elicitation still has lower TV: 0.1937 versus 0.2189, a difference of 0.0252 (12.99%), with interval $[0.0211, 0.0292]$, adjusted t $p = 0.000065$, and adjusted sign $p = 0.125$. Reducing the influence of larger studies retains the conclusion. Secondary complete-table distances agree under both weights (supplement).

The overall result masks unequal gains and losses. Table 2 gives each BN's mean errors and their difference (Δ). Scrum improves from 0.2605 to 0.2334, and SESSI improves slightly. Both Chrome versions, APS, and both log BNs worsen, with the largest increases in the five log nodes. RNM nevertheless wins on 31 of 61 individual nodes. Larger deteriorations outweigh those gains: the aggregate compares error magnitudes, not win counts. BN-level contrasts are descriptive, not seven separate population effects.

Two rows make the differences concrete. Figure 2 compares probability allocations for two configurations not queried by RNM. In panel (a), a Scrum team has Low Self-Organizing but Medium Collaboration. The reference assigns 76.6% probability to Low Cohesion. Direct elicitation assigns 58.0%; RNM assigns 67.6%, closer to the reference. Mean row TV falls from 0.1862 to 0.0901. Thus reconstruction can improve agreement beyond the elicited anchors.

Table 3: All eight conditions: complete-table TV across all 61 nodes, with equal node weights and five replications. Each row pairs the methods under the same prompt and evidence.

Prompt	Evidence	Direct	RNM	Δ
Baseline	None	0.2327	0.2396	+0.0070
Baseline	RAG	0.2349	0.2419	+0.0070
Structured	None	0.2282	0.2389	+0.0107
Structured	RAG	0.2311	0.2413	+0.0102

Panel (b) shows a different outcome. A Chrome component has three or four contributors and no minor contributors. Contributors are classified by their share of changes, so all must be major: the Major Contributors node belongs in the three-to-four category. The reference and all direct responses assign probability one there. RNM assigns 84.4%, spreading 15.6% into neighboring categories. Its reconstruction weakens a count constraint preserved by direct elicitation. These selected examples explain opposing effects without establishing their prevalence across the sample.

Supplementary anchor and fitting diagnostics explore reconstruction compatibility without establishing a single cause of the full-sample result. **RQ1 favors direct elicitation on average under both weighting rules, despite RNM gains in some nodes and BNs.**

4.2 RQ2: Do richer prompts or qualitative evidence help?

Structured prompting offers a limited benefit. Table 3 pairs the methods under identical support. Read across rows to compare direct and RNM, and down each method's column to compare prompts and evidence. Every Δ is positive: direct elicitation remains closer in all four pairs. Its lowest mean TV, 0.2282, occurs with structured prompts and no RAG.

Averaging over evidence conditions, structured prompting lowers direct TV by 0.00413 (interval $[-0.00595, -0.00232]$, Holm-adjusted t $p = 0.0255$). This is the only RQ2 contrast that rejects after eight-test correction. Equal-family weighting gives a similar direction (−0.00508) but an interval crossing zero and adjusted $p = 0.4953$. No RQ2 exact sign test rejects. For RNM, the prompt effect is only −0.00067 and does not reject after adjustment.

Qualitative RAG provides no demonstrated gain. Adding RAG raises mean error under both prompt styles in Table 3. Averaged over styles, the change is +0.00252 for direct elicitation and +0.00230 for RNM; neither adjusted test rejects. The Cohesion card illustrates the information provided: shared goals and joint responsibility, without conditional frequencies. **RQ2 supports structured direct elicitation only under node weighting and demonstrates no benefit from this qualitative RAG.** This does not establish equivalence or rule out benefits from local quantitative evidence. Full contrasts appear in the supplement.

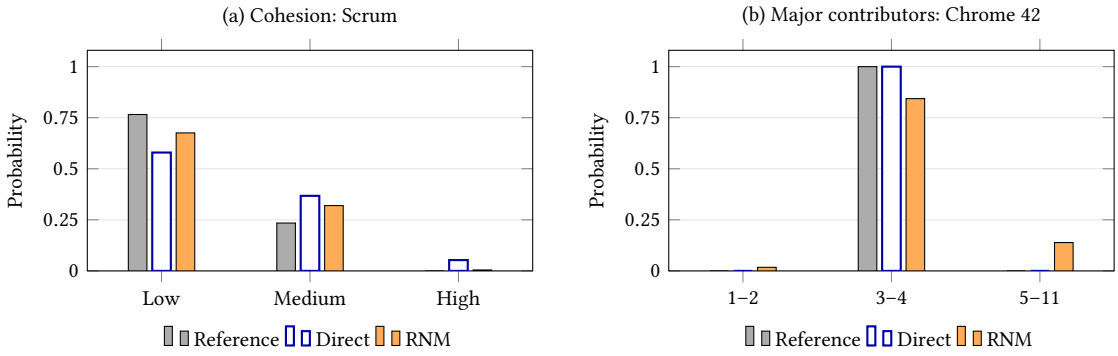


Figure 2: Two CPT rows unqueried by RNM. (a) Cohesion given Low Self-Organizing and Medium Collaboration. (b) Major contributors given three or four total contributors and no minor contributors. Bars average 20 predictions per method. Examples were selected for opposing effects in the earlier non-anchor comparison; they illustrate mechanisms and do not represent the full sample.

Table 4: Scaling for ternary variables. Requested-row reductions are exact for our anchor rule. Total-token savings are conditional projections beyond the observed maximum of 27 rows.

Parents	Rows	Anchors	Exact row reduction	Projected total token reduction
4	81	11	86.4%	36.4%
5	243	13	94.7%	65.9%
8	6,561	19	99.7%	—

4.3 RQ3: Do fewer judgments translate into fewer tokens?

Observed savings depend on the resource measured. RNM requests 5,040 distributions versus 8,420: 40.1% fewer. Output tokens fall from 341,153 to 299,798 (12.12%), whereas input plus output falls from 14,697,090 to 14,607,517 (0.61%). Repeated context and instructions dominate these small-CPT requests. Across five paired replication totals, Holm-adjusted t tests support output savings ($p = 0.000073$), but not total-token savings ($p = 0.0809$); both adjusted exact sign tests give $p = 0.125$. The supplement reports totals, intervals, latency, and fitting time.

Larger tables offer greater elicitation reduction. For ternary parents, full-table elicitation grows as 3^k, while our anchor count grows linearly for $k \geq 3$. Table 4 distinguishes this exact reduction from token scenarios. Among the five observed 27-row CPTs, mean output tokens fall from 823 to 375 (54.5%), and total tokens from 14,372 to 12,750 (11.3%). All five are Scrum, so this descriptive comparison cannot separate size from BN-specific factors.

Token projections are conditional scenarios. The fitted total-token equations are approximately $\widehat{T}_D = 11{,}239 + 118N$ and $\widehat{T}_R = 11{,}197 + 185A$. Under constant fitted costs, 81-row CPTs require about 20,800 direct versus 13,200 RNM tokens; 243-row CPTs require 39,900 versus 13,600. These extrapolations assume the same output format and anchor policy. More parents can increase input length per row, reasoning, repairs, or request splitting; maintaining accuracy may require extra anchors. Full-table generation and storage still require cN entries. **RQ3 establishes fewer judgments and output tokens in this sample; larger-CPT token savings remain conditional, and their accuracy–effort trade-off is untested.**

5 Discussion

The three RQs inform how to select and validate a CPT construction pipeline, considering reference agreement, elicitation support, and token usage.

5.1 Research questions in relation to the literature

RQ1: Reconstruction is an assumption, not an accuracy guarantee. RNM imposes a shared mathematical relationship across rows [11, 15]. This can stabilize uncertain judgments, but also spread error when the intended relationship differs. Scrum's RNM-derived reference makes compatibility plausible; Chrome's contributor-count example exposes a constraint that smoothing weakens. Restricted diagnostics support compatibility as an explanation, without isolating it from influence assumptions, LLM judgments, or optimization. The other BNs show why this interpretation cannot be reduced to a Scrum–Chrome dichotomy.

Blomaard et al. fit semi-automatic methods against known CPTs [2], examining what their representations can approximate. Our fits use uncertain LLM judgments, testing the whole pipeline. Nafar et al.'s gains from finer probability-query granularity [18] concern asking smaller questions, rather than omitting configurations. These comparisons distinguish elicitation quality from reconstruction compatibility: ordinal states make RNM applicable, but do not establish its suitability for every relationship.

The node-weighted difference of 0.0087 is 0.87 percentage points on the TV scale, versus 2.52 under equal-family weighting. Its statistical significance does not establish a consequential difference in SE decisions. These are differences in average reference error, not distances between the methods' predictions or uniform shifts in probabilities. Gains and losses can cancel, and a small change near a

decision threshold may matter. Without an acceptable-error margin or downstream evaluation, practical superiority and equivalence remain unresolved.

RQ2: Instructions and evidence address different problems. Structured prompts clarify conditioning and normalization without supplying missing domain knowledge. Their limited benefit for direct elicitation is consistent with that distinction, although our design does not isolate which instruction helped. For RNM, improved judgments must also survive fitting. Weighting-sensitive findings do not establish a general prompting advantage across SE contexts.

Qualitative RAG reinforces meanings and relationships already partly expressed in the common brief; it cannot uniquely determine conditional frequencies. Gottal and Matthes instead use numerical summaries to constrain reconstruction [12], while Cypko et al. assess clinical modeling support [6]. Our result concerns numerical agreement under a particular evidence design, not RAG's value for modeling assistance. Whether company measurements improve elicitation remains untested.

RQ3: Human burden and token usage are different resources. Partial elicitation reduces expert judgments [1], whereas repeated context can dominate LLM token usage. Our small-CPT sample therefore limits the observed economy. The anchor rule offers increasing elicitation reduction as CPTs grow, but token scenarios do not establish accuracy or end-to-end savings. The current numerical integration uses 5^k points per row; larger tables require computationally scalable fitting and generation as well as validation of fewer judgments. Batching or context reuse could also change both methods' costs.

5.2 Implications for SE researchers and practitioners

For researchers. Evaluate complete CPTs alongside diagnostics separating raw judgments from fitting effects. Report aggregate error and variation across BNs: conclusions can depend on study representation or large deteriorations despite many small gains. Match the endpoint to the claim. BIRD evaluates LLM judgments coupled with BN inference [9]; similarly, SE studies should test whether CPT differences change release, prioritization, or process-improvement decisions. Reference agreement alone cannot demonstrate decision value or explainability.

For practitioners. Use direct elicitation as the initial numerical baseline suggested by this sample; consider RNM when its compensatory or constraining relationships match domain knowledge. Check logical constraints before accepting smooth distributions: contributor counts can exclude otherwise plausible states. Treat either output as a candidate requiring review. Generic definitions and reusable BN components can initialize a model, but organization-specific probabilities require local evidence and validation. Experts could focus on consequential or uncertain relationships, supported by sensitivity analysis and held-out company cases. This is a proposed validation strategy, not evidence that limited expert input necessarily suffices.

6 Threats to Validity

Construct validity. TV measures reference agreement, not company outcomes; equal-row scoring gives rare and common configurations equal influence. Complete-table evaluation and supplementary distances check sensitivity to the endpoint, but cannot establish calibration, explanation quality, or decision utility. RQ1 therefore compares numerical outputs, not expert replacement. No acceptable-error margin establishes practical sufficiency.

Reference validity. Scrum's RNM-derived reference may favor RNM; prompt and fitting development also used that setting. SESSI represents an original elicitation stage. We checked dimensions, normalization, and indexing and reported BN-specific results and fitting diagnostics. These cannot certify Scrum against an independent AgenaRisk export. Shared definitions control presentation, not the correctness of adapted meanings or uncertain influence directions. Cross-BN differences therefore admit multiple explanations.

Internal validity. Fresh sessions, disabled tools, withheld references, and reference-blind fitting limited direct access, not public-BN memorization. Logs record requested settings, not an independently verified serving version. Schema checks and format repairs ensure valid distributions, not semantic fidelity; exhaustive human review was absent. Alternative fits checked sensitivity, but bounded searches can miss better fits and reference-informed diagnostics are optimistic. RQ2 evaluates bundles, preventing attribution to individual instructions or cards.

Conclusion validity. We paired replication blocks rather than treating correlated nodes or rows as independent, reported intervals, and combined Holm-adjusted tests with exact-sign sensitivity checks. Five blocks permit weak checks of normality, independence, and stationarity; conflicting tests limit confidence. The SE restriction, complete-table endpoint, weighting, and tests followed earlier result inspection, making inference exploratory. Separate Holm families do not provide study-wide confirmatory control; non-rejection does not establish equivalence.

External validity. The sample spans five publication families; Scrum and Chrome supply 52 of 61 nodes. Equal-family weighting checks their influence without making the sample representative. CPTs have at most 27 rows. Token projections use only three size levels, with all five largest CPTs from Scrum; size and BN origin are confounded. Linear extrapolation assumes stable costs and does not predict accuracy. We tested only GPT-6 Astra at medium effort and qualitative evidence. Larger CPTs, other implementations, and local numerical evidence require validation.

7 Conclusion

Requesting fewer probabilities from an LLM does not by itself improve the completed CPT. Across 61 ordinal nodes from seven SE BNs, direct elicitation provided the stronger average numerical baseline, while RNM helped some relationships and weakened others. Structured prompting offered limited support, and qualitative RAG added no demonstrated benefit beyond common definitions. Smaller RNM outputs yielded little reduction in total tokens on these small CPTs, despite increasing theoretical elicitation savings at larger sizes. The lesson is to evaluate the completed table and full

interaction before equating fewer judgments with better or cheaper modeling.

Our contribution is a controlled comparison of complete construction pipelines, including anchors, systematically varying prompting and evidence. Network-level results, weighting sensitivity, and fitting diagnostics distinguish the overall tendency from variation between relationships. Statistical differences in reference agreement establish neither practical superiority nor deployment readiness; five replications and test sensitivity limit inference.

The next priority is to test whether organization-specific measurements and targeted expert review improve candidate CPTs and their decisions. Evaluation should measure calibration and decision quality on held-out company cases. A complementary direction is to select direct elicitation or RNM by relationship, using constraints and compatibility checks established without test references. Additional models, larger CPTs, and controlled context reuse would test whether the accuracy–effort trade-off changes. This would move from generating plausible tables toward demonstrating their value for SE decision support.

Data Availability Statement

The anonymous replication package provides prompts, reference CPTs, responses, fitted tables, evaluation code, token counts, and supplementary analyses: https://anonymous.4open.science/r/llm-rnm-se-replication-13FA.

Use of AI Statement

Codex (OpenAI) was used under the author's direction to assist with language revision, organization, consistency checking, and the preparation of LaTeX code for figures and tables.

References

[1] Martine J. Barons, Steven Mascaro, and Anca M. Hanea. 2022. Balancing the Elicitation Burden and the Richness of Expert Input When Quantifying Discrete Bayesian Networks. *Risk Analysis* 42, 6 (2022), 1196–1234. doi:10.1111/risa.13772

[2] Bodille P. M. Blomaard, Gabriela F. Nane, and Anca M. Hanea. 2025. Approaches for Reducing Expert Burden in Bayesian Network Parameterization. *Entropy* 27, 6, Article 579 (2025), 27 pages. doi:10.3390/e27060579

[3] Sicong Cao, Xiaobing Sun, Ratnadira Widyasari, David Lo, Xiaoxue Wu, Lili Bo, Jiale Zhang, Bin Li, Wei Liu, Di Wu, and Yixin Chen. 2025. A Systematic Literature Review on Explainability for ML/DL-based Software Engineering. *Comput. Surveys* 58, 4, Article 95 (2025), 34 pages. doi:10.1145/3763230

[4] Alexander Capstick, Rahul Krishnan, and Payam Barnaghi. 2025. AutoElicit: Using Large Language Models for Expert Prior Elicitation in Predictive Modelling. In *Proceedings of the 42nd International Conference on Machine Learning (Proceedings of Machine Learning Research, Vol. 267)*. PMLR, 6746–6777. https://proceedings.mlr.press/v267/capstick25a.html

[5] Kwai-Sang Chin, Da-Wei Tang, Jian-Bo Yang, Shui Yee Wong, and Hongwei Wang. 2009. Assessing new product development project risk by Bayesian network with a systematic probability generation methodology. *Expert Systems with Applications* 36, 6 (2009), 9879–9890. doi:10.1016/j.eswa.2009.02.019

[6] Mario A. Cypko, Muhammad Agus Salim, Aditya Kumar, Leonard Berliner, Andreas Dietz, Matthaeus Stoehr, and Oliver Amft. 2026. Large language models with retrieval-augmented generation enhance expert modelling of Bayesian network for clinical decision support. *International Journal of Computer Assisted Radiology and Surgery* 21, 2 (2026), 211–222. doi:10.1007/s11548-025-03524-9

[7] Artur d'Avila Garcez and Luis C. Lamb. 2023. Neurosymbolic AI: the 3rd wave. *Artificial Intelligence Review* 56, 11 (2023), 12387–12406. doi:10.1007/s10462-023-10448-w

[8] Santiago del Rey, Silverio Martínez-Fernández, and Antonio Salmerón. 2023. Bayesian Network analysis of software logs for data-driven software maintenance. *IET Software* 17, 3 (2023), 268–286. doi:10.1049/sfw2.12121

[9] Yu Feng, Ben Zhou, Weidong Lin, and Dan Roth. 2025. BIRD: A Trustworthy Bayesian Inference Framework for Large Language Models. In *The Thirteenth International Conference on Learning Representations*. https://openreview.net/forum?id=fAAaT826Vv

[10] Norman Fenton, Martin Neil, William Marsh, Peter Hearty, Łukasz Radliński, and Paul Krause. 2008. On the effectiveness of early life cycle defect prediction with Bayesian Nets. *Empirical Software Engineering* 13 (2008), 499–537. doi:10.1007/s10664-008-9072-x

[11] Norman E. Fenton, Martin Neil, and Jose Galan Caballero. 2007. Using Ranked Nodes to Model Qualitative Judgments in Bayesian Networks. *IEEE Transactions on Knowledge and Data Engineering* 19, 10 (2007), 1420–1432. doi:10.1109/TKDE.2007.1073

[12] Jonas Gottal and Florian Matthes. 2026. Verifiable Parameterization of Bayesian Networks from Scientific Literature: Unlocking Unstructured Empirical Evidence. In *Findings of the Association for Computational Linguistics: ACL 2026*. Association for Computational Linguistics, 3400–3414. doi:10.18653/v1/2026.findings-acl.167

[13] Jan-Phillip Herrmann, Sven Tackenberg, Elio Padoano, Jörg Hartlief, Jens Rautenstengel, Christine Loeser, and Jörg Böhme. 2022. An ERP Data Quality Assessment Framework for the Implementation of an APS system using Bayesian Networks. *Procedia Computer Science* 200 (2022), 194–204. doi:10.1016/j.procs.2022.01.218

[14] Andrey Krutauz, Tapajit Dey, Peter C. Rigby, and Audris Mockus. 2020. Do code review measures explain the incidence of post-release defects? Case study replications and Bayesian networks. *Empirical Software Engineering* 25 (2020), 3323–3356. doi:10.1007/s10664-020-09837-4

[15] Pekka Laitila and Kai Virtanen. 2020. On Theoretical Principle and Practical Applicability of Ranked Nodes Method for Constructing Conditional Probability Tables of Bayesian Networks. *IEEE Transactions on Systems, Man, and Cybernetics: Systems* 50, 5 (2020), 1943–1955. doi:10.1109/TSMC.2018.2792058

[16] Martí Manzano, Claudia Ayala, Cristina Gómez, Antonin Abherve, Xavier Franch, and Emilia Mendes. 2021. A Method to Estimate Software Strategic Indicators in Software Development: An Industrial Application. *Information and Software Technology* 129 (2021), 106433. doi:10.1016/j.infsof.2020.106433

[17] Emilia Mendes. 2012. Using knowledge elicitation to improve Web effort estimation: Lessons from six industrial case studies. In *Proceedings of the 34th International Conference on Software Engineering*. IEEE, 1112–1121. doi:10.1109/ICSE.2012.6227108

[18] Aliakbar Nafar, Kristen Brent Venable, Zijun Cui, and Parisa Kordjamshidi. 2026. Extracting Probabilistic Knowledge from Large Language Models for Bayesian Network Parameterization. *Transactions on Machine Learning Research* (May 2026). https://openreview.net/forum?id=Fy3Byg3CVo

[19] Mirko Perkusich, Kyller Gorgonio, Hyggo Almeida, and Angelo Perkusich. 2017. A Framework to Build Bayesian Networks to Assess Scrum-based Software Development Methods. In *Proceedings of the 29th International Conference on Software Engineering and Knowledge Engineering*. 7 pages. doi:10.18293/SEKE2017-139

[20] Mirko Perkusich, João Nunes, Emilia Mendes, Emanuel Dantas, Ademar Sousa, Danyllo Albuquerque, Kyller C. Gorgônio, and Angelo Perkusich. 2026. Semi-Automatic Quantification of Bayesian Networks for Software Decision Support: Comparing WSA and RNM in a Software R&D Organization. *arXiv preprint arXiv:2609.31275* (2026), 37 pages. https://arxiv.org/abs/2609.31275

[21] Mirko Perkusich, Gustavo Soares, Hyggo Almeida, and Angelo Perkusich. 2015. A procedure to detect problems of processes in software development projects using Bayesian networks. *Expert Systems with Applications* 42, 1 (2015), 437–450. doi:10.1016/j.eswa.2014.08.015

[22] Kumar Rahul and Shovan Chowdhury. 2026. Human–AI Construction of Bayesian Networks for Operational Decision Support: A Virtual Survey Approach. *arXiv preprint arXiv:2607.14141* (2026). doi:10.48550/arXiv.2607.14141

[23] Thiago Rique, Mirko Perkusich, Emanuel Dantas, Danyllo Albuquerque, Kyller Gorgônio, Hyggo Almeida, and Angelo Perkusich. 2023. On Adopting Software Analytics for Managerial Decision-Making: A Practitioner's Perspective. *IEEE Access* 11 (2023), 73145–73163. doi:10.1109/ACCESS.2023.3294823

[24] Thiago Rique, Mirko Perkusich, Kyller Gorgônio, Hyggo Almeida, and Angelo Perkusich. 2025. Constructing the graphical structure of expert-based Bayesian networks in the context of software engineering: A systematic mapping study. *Information and Software Technology* 177 (2025), 107586. doi:10.1016/j.infsof.2024.107586